Through the Lens:
4 Years of Campus Life in 57k Photos

Milestone 1 – COM-480

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1 Dataset

Our dataset was fully built in-house rather than derived from a public benchmark. It contains metadata from 56,890 photographs shot over four years on the EPFL campus by one team member (Adrien Buttier), who regularly documented both everyday student life and major events. This gives the dataset a strong real-world coherence: a consistent ecosystem, a long time span, and stable photographic practices. The extracted table (about 190 MB, no full-resolution pixels in the base file) includes 952 metadata fields, covering both capture-time camera information and post-processing traces.

Instead of relying on a single feature type, the dataset combines several complementary families: temporal and organizational context (dates, albums, directory structure), camera acquisition parameters (exposure time, aperture, ISO, shutter behavior), camera/lens descriptors, file and geometry descriptors, and Lightroom/Camera Raw editing metadata. Beyond these native fields, we also computed visual descriptors by generating RGB histograms (before/after rendering), which adds a compact representation of color distributions for large-scale comparison.

Overall quality is good for the dimensions most relevant to visualization and exploratory analysis, especially technical and temporal signals. As a result, preprocessing is expected to be moderate: mostly feature selection, type/unit harmonization, handling missingness, and grouping high-dimensional metadata into coherent analysis blocks, rather than heavy correction of broken records.

A potential extension is to generate ultra-small image proxies ($\sim$250 px wide), so the image layer can fit in roughly 3 GB and support interactive hover previews while naturally anonymizing most people through low resolution. We also plan to add a lightweight set of semantic tags to improve comparability across scenes.

2 Problematic

What are we trying to show? We reveal the hidden story behind four years of campus photography through metadata, the data cameras and editing software silently recording. Rather than analyzing the photos themselves, we turn invisible technical information into visual narratives, reconstructing campus life as seen through Adrien's lens.

Motivation. Every photograph carries a rich but invisible layer of information: camera settings, time of day, lens choice, and post-processing decisions. Individually, these numbers mean little. Aggregated across four years, they expose meaningful patterns about campus life, shooting conditions, and stylistic evolution.

Main axes of exploration:

1. **Temporal patterns.** How does photographic activity evolve over time? Calendar heatmaps and hourly charts reveal the rhythms of campus life: major events, semester peaks, and quiet periods.

2. **Technical fingerprint.** ISO, aperture, and shutter speed form the exposure triangle. Interactive scatter plots with brushing and linking let users explore how these parameters relate to each other and to different shooting contexts.

3. **Creative evolution.** Lightroom editing records reveal how post-processing style develops over time. Ridgeline plots track shifts in key sliders (Clarity, Shadows, Dehaze, Vibrance) year over year, complemented by a radar chart comparing editing fingerprints across periods.

4. **Gear story.** Camera and lens records trace the progression of equipment and its influence on creative output, visualized through Sankey diagrams and stacked area charts.

Incorporating the actual photos. Thumbnails appear on hover over data points, grounding the metadata in reality: a high-ISO data point becomes meaningful when you can see it was a dimly lit party photo.

Target audience. Students curious about campus culture, photography enthusiasts, and anyone interested in how metadata can unlock large visual archives.

3 Exploratory Data Analysis

This section explores the dataset across four perspectives: temporal patterns, photographic settings, equipment usage, and post-processing adjustments.

3.1 Data Completeness

We filter the original 952 metadata fields, retaining only those with at least 90% coverage, reducing the feature space to 326 columns. Core fields such as `DateTimeOriginal`, `ISO`, `FNumber`, and `Model` are present for over 99% of images. Table 1 breaks down completeness levels, confirming that retained fields provide reliable signal for analysis.

Completeness Category	Columns	Description
Perfect (100%)	9	No missing values
Nearly perfect (99–99.9%)	199	Only a few records missing
Excellent (95–98.9%)	32	Very high coverage
Good (90–94.9%)	82	Minor missing data but still reliable

Table 1: Summary of metadata completeness across the 326 analyzed columns.

3.2 Temporal Distribution

Photographic activity follows clear temporal patterns (Figure 1). Yearly totals peak in 2024, while monthly aggregation highlights strong activity in May and September, corresponding to major campus events and the start of the academic year. The hourly distribution is bimodal, with peaks around midday and early evening, reflecting academic and social rhythms.

3.3 Exposure Statistics

Exposure metadata reveals how photographs were captured (Figure 2). Most images use low ISO values typical of well-lit environments, while higher ISO values correspond to darker scenes such as indoor events. Apertures concentrate around wide settings (e.g., f/2.8), and shutter speeds cluster around typical handheld values ($\sim$1/125 s).

3.4 Gear Usage

Camera usage is highly concentrated (Figure 3). The Sony ILCE-7M4 accounts for most images, providing technical consistency across the dataset. Lens metadata required partial recovery for fixed-lens cameras such as the Fujifilm X100VI and Panasonic DMC-LX100. After recovery, 53 lenses were identified, with usage concentrated around a few frequently used Sony lenses.

3.5 Lightroom Editing Patterns

Post-processing metadata reveals consistent editing behavior (Figure 4). Clarity adjustments are generally small, dehaze is applied selectively, and shadow values

Figure 1: Temporal distribution of photographs across years, months, and hours, highlighting peaks during major campus events and evening activity.

are strongly positive, indicating a workflow focused on improving visibility and recovering detail in darker areas of the image.

4 Related Work

Existing work with EXIF data. Tools like PixelPeeper, ExifTool, and Flickr's aggregate stats enable per-photo or community-level metadata browsing. The academic project *PhotoVis* (UC Berkeley, 2011) added interactive filtering over aggregate EXIF data. However, none trace the longitudinal story of a single photographer's evolving practice.

Why our approach is original. Three things distinguish our project. First, the dataset is personal and curated: it captures one team member's four-year documentation of EPFL, with direct contextual knowledge of every album. Second, we combine EXIF capture metadata with Lightroom/XMP editing metadata, which are rarely analyzed together—letting us visualize not just how photos were taken, but how they were developed. Third, we frame the work as a narrative about campus life and creative evolution, not a technical photography tool.

Sources of inspiration. Two pieces shaped our thinking. The Pudding's

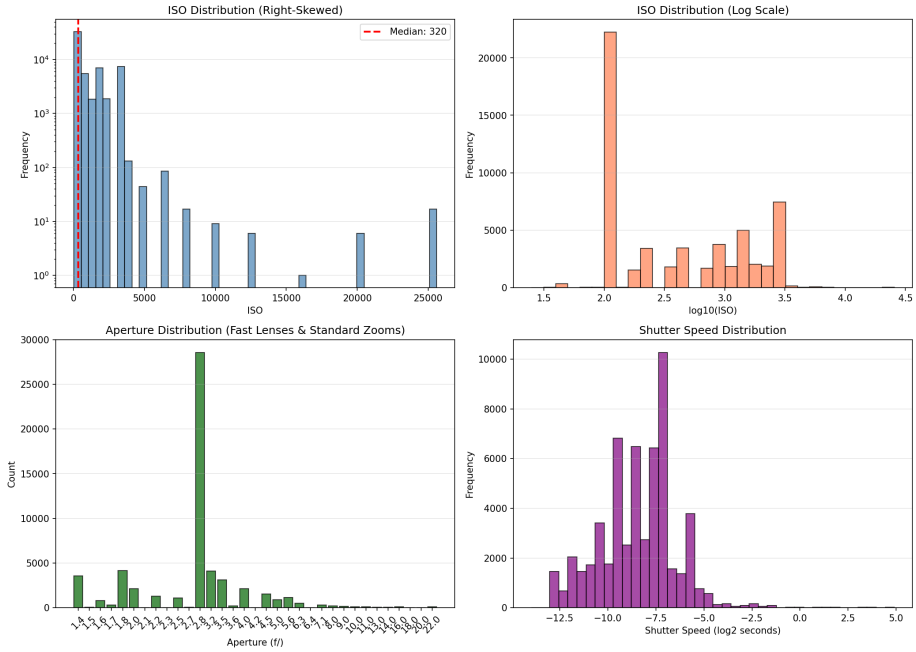

Figure 2: Distribution of exposure parameters. Most photos use low ISO, wide apertures, and shutter speeds typical of handheld event photography.

"Is the Love Song Dying?"[1] plots 5,000+ Billboard hits as color-coded bubbles across decades, letting you watch popular music's palette shift while still hovering on individual songs. We borrow this structure: each of Adrien's 57,000 photos as a dot, laid out chronologically and coded by lens or context, so viewers can trace his photographic evolution at any scale (Figure 5).

Nathan Yau's "A Day in the Life of Americans"[2] animates 1,000 dots across a simulated day. The metaphor differs—a photo has fixed settings—but the insight translates: sort 57,000 photos by hour and a day/night rhythm emerges; sort by ISO and indoor parties separate from outdoor festivals. The feeling of collective behavior emerging from individual dots is exactly what we're chasing.

This dataset has not been used in any other course or project.

[1] `https://pudding.cool/2024/11/love-songs`
[2] `https://flowingdata.com/2015/12/15/a-day-in-the-life-of-americans/`

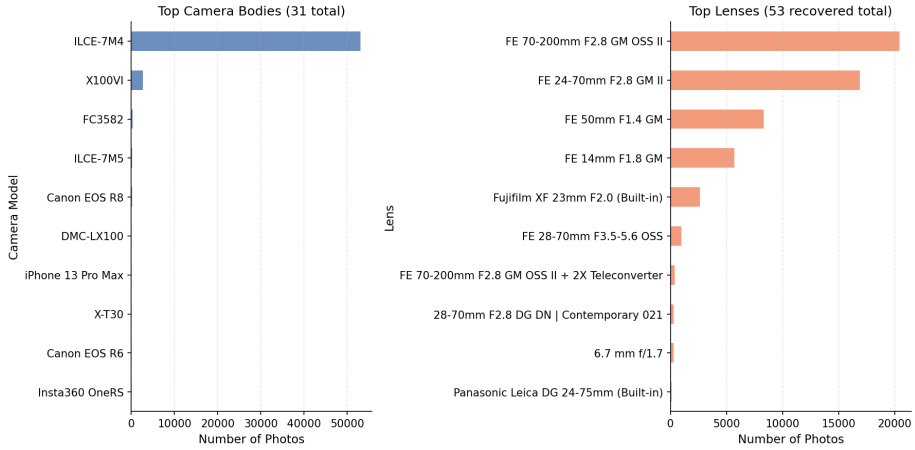

Figure 3: Camera and lens usage in the dataset. Most photographs were captured with the Sony ILCE-7M4, while lens usage concentrates on a few frequently used lenses.

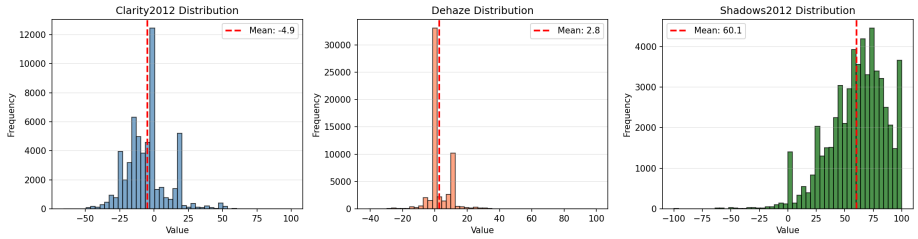

Figure 4: Distributions of Lightroom editing parameters. Clarity changes are small, dehaze is selective, and shadows are frequently increased to recover detail.

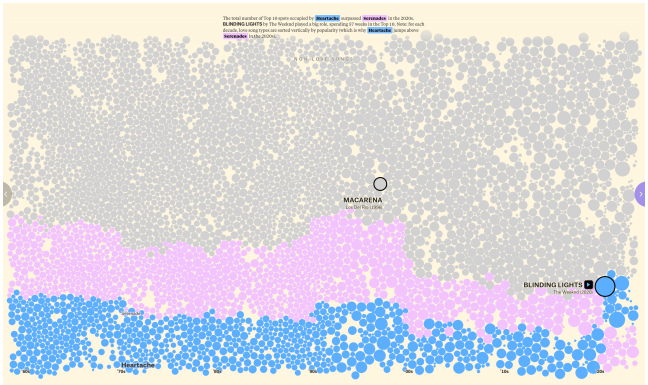

Figure 5: The Pudding's "Is the Love Song Dying?"—our primary visual inspiration. Each bubble is a song; we plan a similar approach where each bubble is a photo.